Chapter 5

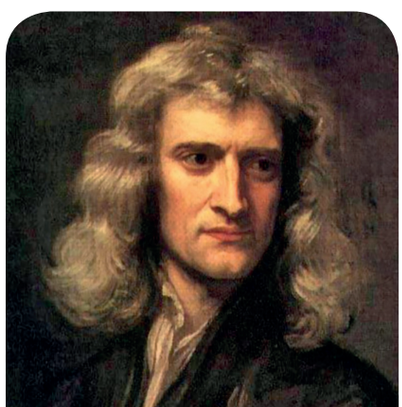
DIFFERENTIAL CALCULUS

Learning Objectives

After studying this chapter, the students will be able to understand

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- the concept of limit continuity
- formula of limit and definition of continuity
- basic concept of derivative
- how to find derivative of a function by first principle
- derivative of a function by certain direct formula and its respective application
- how to find higher order derivatives



Isaac Newton



G.W. Leibnitz

Introduction

Calculus is a Latin word which means that *Pebble or a small* stone used for calculation. The word calculation is also derived from the same Latin word. Calculus is primary mathematical tool for dealing with change. The concept of derivative is the basic tool in science of calculus. Calculus is essentially concerned with the rate of change of dependent variable with respect to an independent variable. Sir Issac Newton (1642 – 1727 CE) and the German mathematician G.W. Leibnitz (1646 – 1716 CE) invented and developed the subject independently and almost simultaneously.

In this chapter we will study about functions and their graphs, limits, derivatives and differentiation technique.

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